

COMP532-202425 Assignment 2

You need to solve each of the following problems. The assignment aims to design and implement a deep reinforcement learning agent for a video game from OpenAI Gym or Gymnasium. You must also include a brief report describing and discussing your solutions to the problems. Students can do the assignment in groups or individuals.

- This assignment is worth 15% of the total mark for COMP532
- 80% of the assignment marks will be awarded for correctness of results
- 20% of the assignment marks will be awarded for the quality of the accompanying report
- Students will do the assignment in groups
- The assignment marks will be awarded for correctness of results
- We expect 5 students in one group (it would be fine to have groups of 1, 2, 3, and 4 as well, but it is suggested to have groups of 5), please find your team members on your own.
- Every student needs to submit the report.
- The same marks will be granted to all the members in the same group
- Please list all your group members (names, emails, student ids) and individual contributions in your submitted report

Submission Instructions

- **Deadline: See online system.**
- Send all solutions as a single PDF document containing your answers, results, and discussion of the results. Attach the source code for the programming problems as separate files.
- Submit your solution via Canvas.
- Penalties for late submission apply in accordance with departmental policy as set out in the student handbook, which can be found at <https://intranet.csc.liv.ac.uk/student/msc-handbook.pdf> and the University Code of Practice on Assessment, found at https://www.liverpool.ac.uk/media/livacuk/tqsd/code-of-practice-on-assessment/code_of_practice_on_assessment.pdf

Problem 1 (80 marks)

Implement a deep reinforcement learning agent for a game or environment of OpenAI Gym or Gymnasium.

Use the lunar_lander environment:

https://gymnasium.farama.org/environments/box2d/lunar_lander/.

Please plot the learning progress of your method from 0 to 1000 episodes. You can have a figure to show rewards and another figure to show training loss.

Please use a video or gifs or figures to demonstrate how your agent works.

Prepare a report explaining your solution and containing your results, and discussion of the results.

Attach the source code as separate files. For example, .ipnb - an ipython notebook file.

Problem 2 (20 marks)

Explain exploration and exploitation for deep reinforcement learning.